



磁北科技
MAGNETIC NORTH

MDVR Z8

User Manual



Catalogue

1、Product introduction	3
1.1、Product Overview	3
1.2、Product appearance and specification parameters	3
1.2.1 Product Appearance	3
1.2.2 Specification parameters	4
1.3、Product features, functions and applications	5
1.3.1 Product Features	5
1.3.2 Product Functions	6
1.3.3 Product Applications	9
1.4、Exterior dimensions and panel introduction	10
1.4.1 Exterior Dimensions	10
1.4.2 Front Panel Introduction	11
1.4.3 Rear Panel Introduction	12
1.4.4 Wiring Diagram	14
2、Product Installation	15
2.1、Unpacking and accessory inspection	15
2.1.1 unpacking	15
2.1.2 Accessory Inspection	15
2.2、Installation Precautions	18
2.3、Power cord connection	22
2.4、Hard drive loading and precautions	22
2.4.1 caveat	22
2.4.2 Installation	23
2.5、SD card and SIM card loading and cautions	24
2.5.1 caveat	24
2.5.2 Installation method (as shown)	24
2.6、Device Configuration Enabling Functions	24
3、Fault resolution	25
3.1、Device does not start	25
3.2、The device is always rebooting	25
3.3、No video recording of equipment	25
3.4、Video files without sound	26
3.5、Device dialing unsuccessful	26
3.6、Abnormal equipment positioning	26

1、Product introductions

1.1、Product Overview

The Z8 integrates satellite positioning and monitoring, 8-channel 1080P HD video surveillance, 4G wireless communication, remote dispatch, emergency alarm, voice broadcast, LED/LCD display control, and rich expansion interface functions in one.

It features a multi-core processor and powerful neural network engine, supporting various AI smart functions such as passenger flow statistics, DMS, and ADAS. The intelligent driving active safety defense system integrates scheduling, management, monitoring, and passenger flow statistics, combining two key active safety functions: driver behavior analysis (DMS) and assisted driving warning (ADAS). The front-view ADAS camera can monitor the vehicle's front dynamics in real-time, such as potential collision risks, too-close distance, height limit signs, and other critical situations, reminding the driver with warning tones to pay attention and effectively avoid accidents. It is widely used in commercial vehicles such as buses, passenger cars, and long-distance trucks.

1.2、Product appearance and specification parameters

1.2.1 Product appearance



1.2.2 specification parameters

Item		Description
System	OS	Linux
	CPU	ARM CortexA53 quad cores CPU
	DDR	1GB
	Flash	256MB
Audio and video module	Encoding format	H.264/H.265
	Video input	8 ch 1080P HD Video Level: 1Vpp Impedence: 75Ω NTSC/PAL optional
	Video output	1 ch CVBS video output, impedance 75Ω, 1.0Vp-p
	IP Voice	Supports IP intercom between the terminal and the backend.
	Audio input	8 ch audio, 2.0Vp-p
Positioning module	positioning mode	Beidou/GPS/Global Navigation Satellite Positioning
Wireless communication module	3G/4G	Support GPRS/TD-SCDMA/WCDMA/EVDO/FDD-LTE/TDD-LTE
	Antenna interface	Purple FAKRA-Dmale, impedance: 50Ω
	SIM card	1* standard SIM card
WIFI module	WIFI module	IEEE 802.11 b/g/n,2.4GHz; white FAKRA-B male
Storage module	HDD storage	2T (MAX)
	SDD storage	2T (MAX)
	SD storage	256GB(MAX)
Power module	Operating voltage	DC: +9V ~ +36V, typical 12V/24V
	Power consumption	<10W (no external connection), standby power ≤5W
Sensor	3 axis sensor	Support
	Infrared Sensor	1 built-in, supports external sensors
AI	ADAS	1ch, FCW (forward collision warning)
	DMS	1ch, driver fatigue management
	APC	Dual-Lens Camera
Expansion interface	RS232	1 ch
	RS485	1 ch

	Digital input	6 port H (range: 6~30V), 1port L (range: 0~1V)
	USB	USB 2.0A communication interface
	Ethernet	1 Six-Pin Aviation Connector (100Mbps)
Others	Operating temperature	-20°C~+70°C
	Humidity	0~95% (non-condensing)

1.3、 Product features, functions and applications

1.3.1 Product Features

- Support 8+1ch HD 1080P audio and video monitoring and H.265 high compression;
- Support AI technology, integrated with ADAS/DMS/APC;
- Support 256G SD card and a hard disk, with hard disk protection mechanism, hard drive protection for continuous recording in shocking environments;
- Support 4G/3G/2G all network ;
- Support WIFI, Ethernet;
- Support GPS/BeiDou;
- Support ITMS platform ; Including emergency alarm, over-speed alarm, route deviation alarm and equipment malfunction alarm.
- Supports automatic control of LED/LCD display content based on the vehicle's location.
- Supports automatic voice announcements through the vehicle's speakers based on the vehicle's location.
- Support external driver display (MADT);

- Supports infrared remote control.

3.2 Product Function

Category	Function	Function description
Position monitoring	Satellite positioning	Support GPS/BeiDou, Support electronic fence, mileage statistics, inflection point supplement, blind area supplement, track storage and other functions;
Communication	Wireless communication	Support 4G all netcom and dual SIM card;
Video surveillance	Support audio and video	Support 8+1ch HD 1080P audio and video monitoring and H.265 high compression; Support 8 audio input, ADPCM audio coding mode;
	Supports OSD information superposition	Support video overlay license plate number, channel number, time, latitude and longitude, speed, line number and other subtitle information;
	Support dual data	Dual data rate technology is adopted, high data rate is used for local storage, low data rate is used for network transmission

	rate	(adaptive data rate according to network conditions), both local storage and remote network transmission;
	Support back-up exit door image	According to the status of the vehicle port line, the video output can be switched to reverse or exit the door video screen;
Intelligent dispatch	Voice function	Built-in high-quality TTS voice broadcast function, send SMS message through TTS voice broadcast function, car hands-free phone function, IP voice function, remote monitoring function;
	Alarm function	Emergency alarm, parking timeout alarm, overspeed alarm, overspeed warning, off-line alarm, power failure alarm, equipment failure alarm and other alarm functions;
Public transport application	Bus stop announcement	Voice manual/automatic station (support multi-language station);
	LED/LCD screen linkage	Real-time display to the station, warm tips and other information;
	Service terms	With zebra crossing, turning and other road conditions prompt function;
Storage function	HDD	It is suitable for high definition video recording with a maximum of 1*1TB HDD storage.
	SD card	Supports 1 SD card for audio and video data storage;
Human-computer interaction	external driver display	Support external driver display,data display, driver attendance and terminal Settings;
AI	DSM	The first channel connects to a DSM camera for monitoring driver behaviors such as drowsiness, smoking, and phone use, with alarm notifications.
	ADAS	The second channel connects to an ADAS camera for monitoring driving

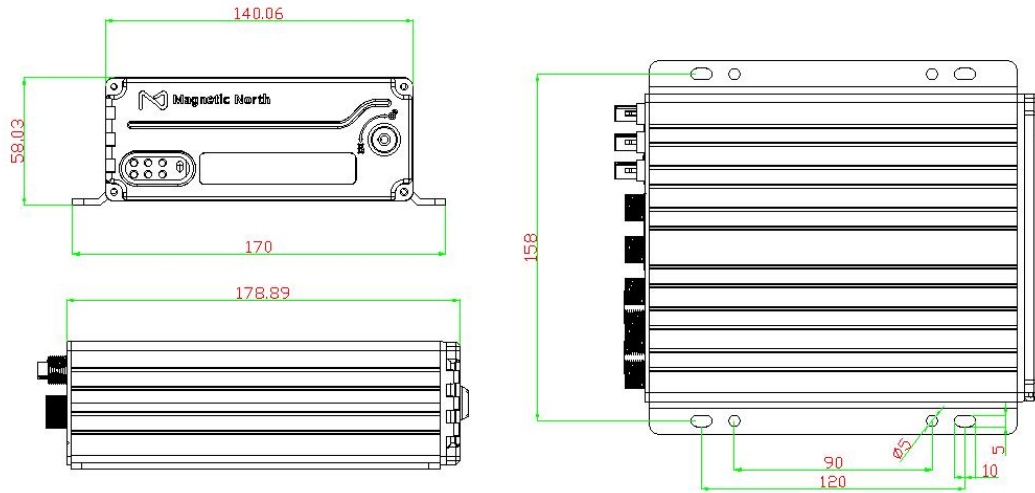
		behaviors such as lane departure and collision, with alarm notifications.
	Passenger Flow	Supports passenger flow statistics for 4 doors.
Device Maintenance	Local Maintenance	Perform program upgrades, route updates, and configuration file updates via a local USB drive.
	Remote Maintenance	Supports scheduled automatic synchronization of server route file information.
Protection	Power Saving Mode	Automatically enters power-saving mode after the vehicle is turned off, configurable to enter power-saving mode after ACC is disconnected for a certain period.
	Delayed Shutdown	Built-in super capacitor protects data of video files before power-off; provides 5-6 seconds of continued operation during sudden power loss.
	Wide Voltage Power Supply	Supports wide voltage input, suitable for 12V and 24V vehicles.

3.3 Product applications

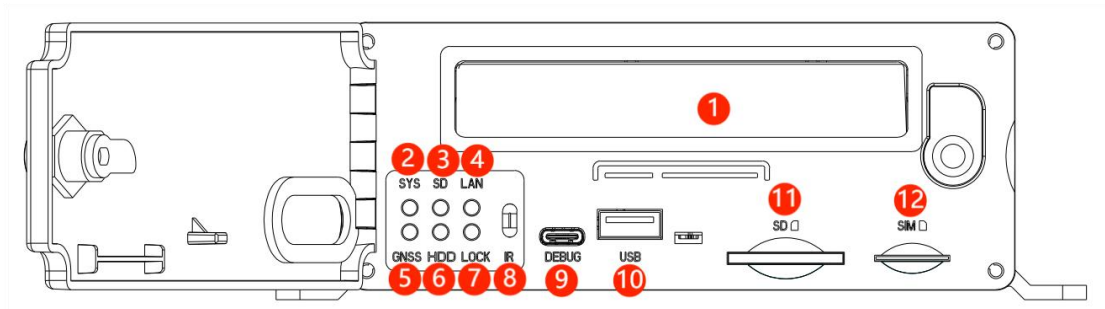


4、Exterior Dimensions and panel introduction

4.1 Exterior Dimensions



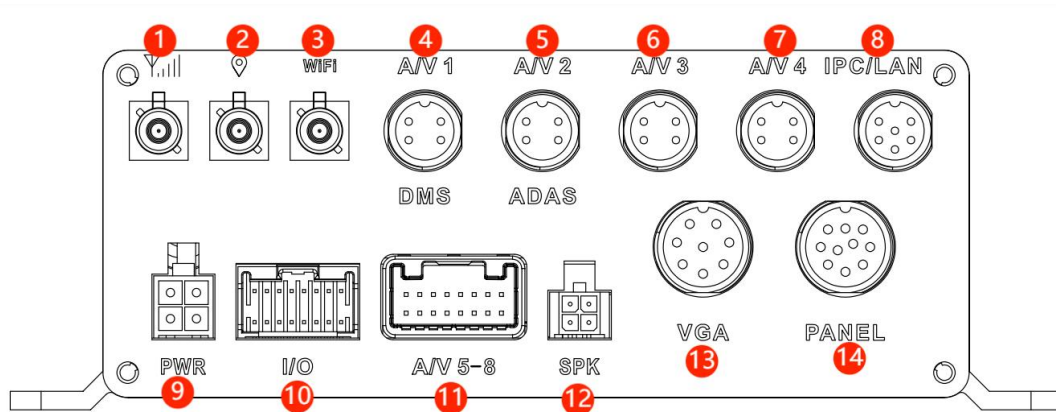
4.2 Front Panel Introduction



No.	silkscreen	introduction
1	HDD	Hard Drive Mounting Location
2	SYS	Indicates device power-up and wireless network connection status Flash: device registration in the network Slow flash: device is registered on the network
3	SD	SD card indicator Light off: storage media not recognized Light on: storage media recognized Light flash: indicates that the storage media is in read/write state
4	LAN	Ethernet indicator Light off: Ethernet not connected Upper Light on: Ethernet connected Light flash: Ethernet has data transmitting and receiving
5	GNSS	Indicates GPS/Beidou positioning signal reception status Off: equipment not positioned on: The device has been located
6	HDD	Hard disc indicator (located on the lower level of the hard disc case) Light off: Storage media not recognized Light on: Recognition to storage media Light flash: Indicates that the storage medium is in the read/write state
7	LOCK	Light on: HDD lock on Light off: HDD lock out
8	IR	infrared receiver

9	DEBUG	CPU Serial Port Printing Information
10	USB	Local operation and maintenance, copying data
11	SD	SD card slot
12	SIM	SIM card slot

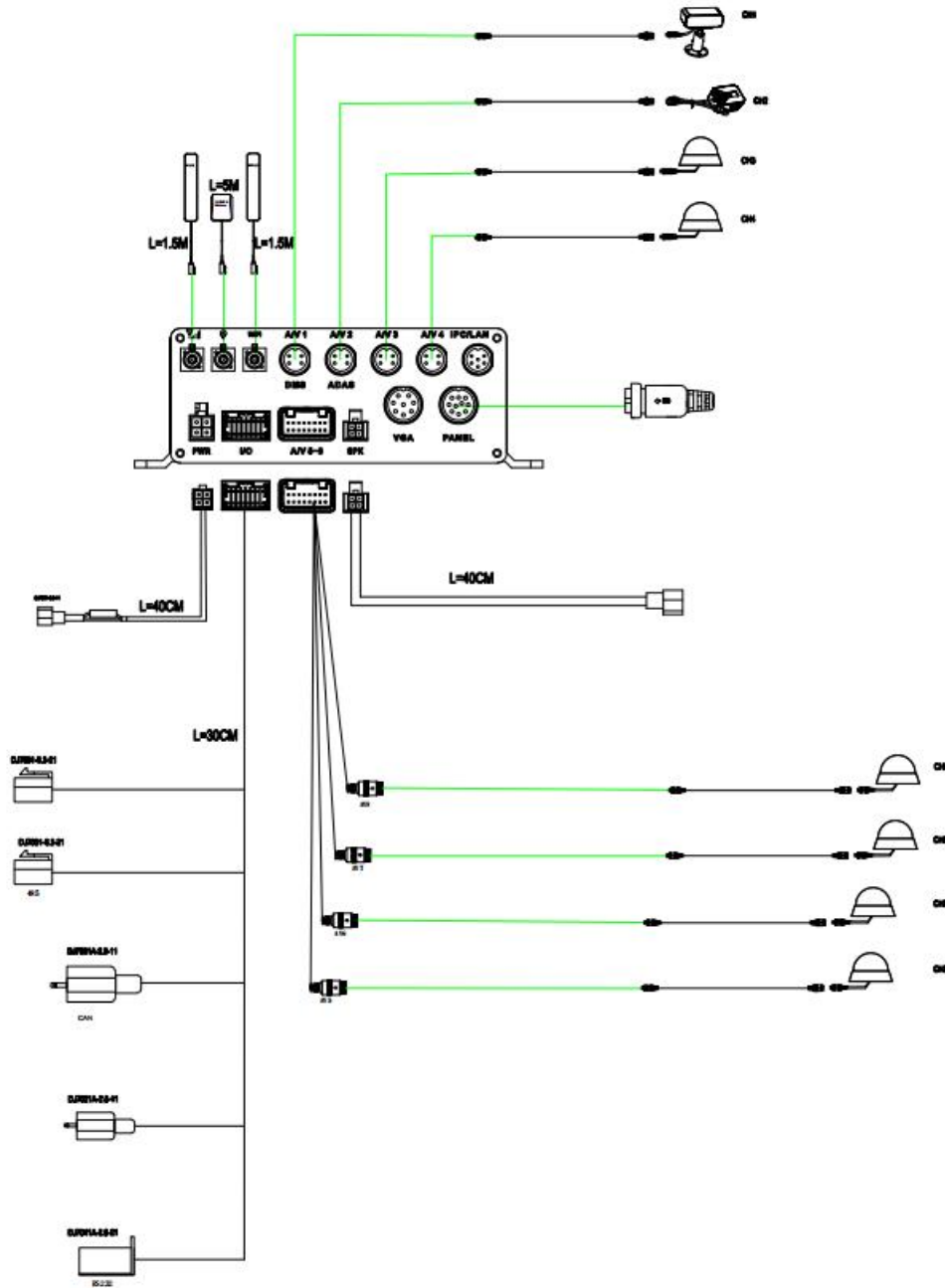
4.3 Rear Panel Introduction



NO.	silkscreen	introduction
1	4G	Brown, for connection to 4G communication antenna
2	GPS	Blue, For connection to GPS/BD positioning antenna
3	WIFI	White ,for connection to WIFI antenna (optional)
4	A/V1 DSM	Connect to AHD/DSM camera 1
5	A/V2 ADAS	Connect to AHD/ADAS camera 2
6	A/V3	Connect to AHD camera 3
7	A/V4	Connect to AHD camera 4
8	IPC/LAN	Ethernet interface
9	PWR	Device power input and ACC input
10	I/O	RS232 interface, RS485 interface, CAN interface, high and low level detection

		interface;
11	A/V 5-8	Connect AHD camera 5-8
12	SPK	Audio output interface for internal and external sound of annunciation
13	VGA	VGA High Definition Screen Output
14	PANEL	Operation screen power supply communication, connecting TTS small speaker, shouting machine, CVBS video output, etc;

4.4 Wiring Diagram



2、Product Installation

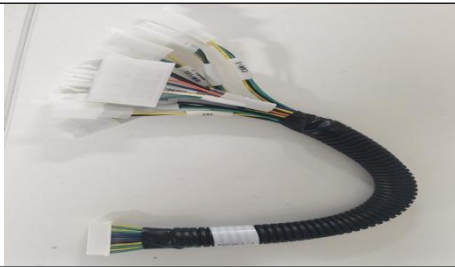
2.1、Unpacking and accessory inspection

2.1.1 Unpacking

Due to the transport process may be bumped and other uncontrollable reasons, if you open the box after the appearance of the host deformation or accessories and other damage, please stop using and contact your supplier;

2.1.2 Accessory inspection

After opening the box, please check whether the accessories are complete in time, the specific accessory information should be with the actual order configuration shall prevail, Z8 vehicle-mounted intelligent terminal factory accessories are as follows:

Name	Note
Standard Accessories	
MDVR Z8	
16 pole adapter cable	
16-pole video communication cable	

4G External Antenna		
WIFI antenna		
BeiDou/GPS external antenna		
4-pole power cord		
4-pole internal and external audio cable		
Terminal' s Key		
certificate of conformity		

Optional accessories		
camera		
ADAS camera		
DSM camera		
4 pole video extension cable		
Infrared remote control equipment		

2.2、Installation Precautions

2.2.1 Note:

In order to ensure the safe use of the Z8 vehicle-mounted intelligent terminal and to obtain satisfactory performance and prolong the service life of the equipment, customers are kindly requested to give full consideration to the following factors when installing the equipment:

- 1) Comply with all electronic product specifications, as well as vehicle and other connected equipment requirements when installing and operating equipment.
- 2) Power and Grounding:
 - a) The equipment uses DC power supply, please pay attention to the positive and negative polarity of the power supply when connecting the power supply.
 - b) The input voltage range of the device is 9V to 36V, please do not connect to a power supply beyond this range. Too low a voltage will result in the in-vehicle equipment not working properly, and too high a voltage will damage the device.
 - c) It is recommended that the device be connected directly to the vehicle's battery power output. Please note that do not connect to the generator power output on the vehicle, as the generator output may generate transient high voltage during start-up and damage the device. Even if the equipment is switched off, the machine is charged inside and short circuits are to be avoided. Disconnect the device from the power supply before connecting other external equipment.

- d) The device's native external output voltage is 12V and is only used to power the camera, do not attach any devices that are not specified as permitted for use on the unit.
 - e) The sensor input method of the device is a level method, the external voltage is considered to be low when it is less than 6V, and is considered to be high when it is in the range of 6V to 30V, and exceeding 30V will result in damage to the device.
 - f) Correctly connect the ground of the equipment to the ground of the vehicle to form a circuit.
 - g) If the machine is not to be used for a long period of time, it is advisable to disconnect the unit completely to prolong its life.
 - h) It is recommended to put wear-resistant, heat-resistant, waterproof and oil-proof sleeve on the outside of the power cord to prevent the line from short-circuiting or breaking due to long-term vibration and friction in the car.
 - i) A 10 amp fuse box is required on the positive end of the power cord near the power output of the vehicle. This prevents the vehicle power supply from burning out if a short circuit occurs in the power line.
 - j) Please use the standard car circuit wiring method to connect the wiring process.
- 3) Humidity requirements:
- a) Install the unit in a dry environment, avoiding locations where moisture, dripping, spraying, etc. is present. Do not install the equipment in a place where water will accumulate in depressions or in a wet place where liquids will drip.

- b) Do not touch the unit with wet hands or while standing in water or in contact with other water sources, there is a risk of electric shock.
- 4) mounting position:
- a) To prolong the life of the device, install the device as far as possible in an area of the vehicle where vibration is weak, such as behind the driver's seat.
 - b) The equipment should be installed in a ventilated part of the vehicle: when installing the device, the front cover needs to reserve 15cm of operating space so that the front cover can be opened for use (as shown in the figure), and the device needs to dissipate heat, so it is not recommended to install it in a narrow confined space.



- c) Host four corners of the fixed screws: it is recommended to use M4 self-tapping screws.
- d) Ensure that the equipment is kept away from heat sources on the vehicle, and that there is no debris piled up around the equipment.
- e) The equipment can only be installed horizontally or sideways vertically (if there is a need for installation in other directions, please consult our technical staff first), any other angle of installation may damage the

equipment and is strictly prohibited.

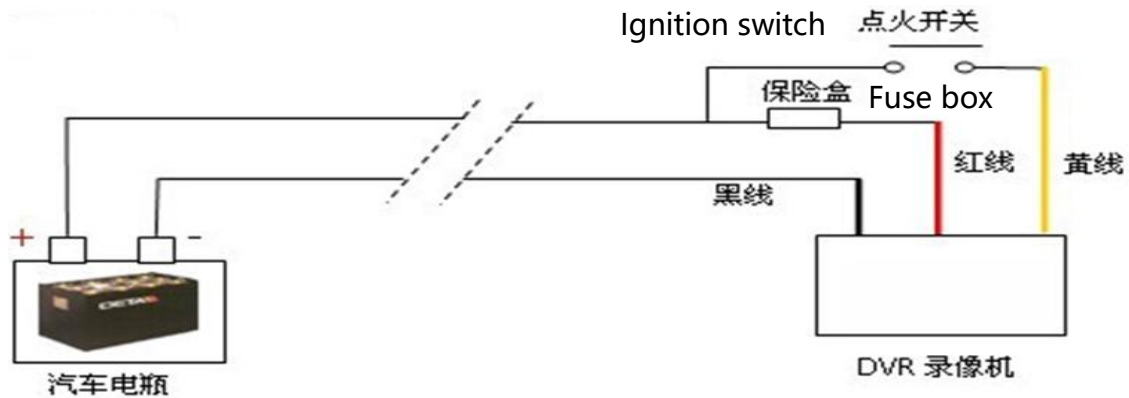
- f) Positioning antenna signals are easily shielded by concrete and metal substances, so when fixing it, make sure that there are no metal substances above its signal receiving surface to block it, and make sure that the receiving surface can be turned upwards towards the sky. It is recommended to install it on the roof of the car, and the antenna is sealed and fixed with glass adhesive around the antenna body and the crossing holes to prevent water seepage.
- g) The WiFi antenna and communication antenna should be avoided to be installed in the space enclosed by metal to prevent the monitoring host from not being able to receive the signal or poor signal.

5) Equipment Safety:

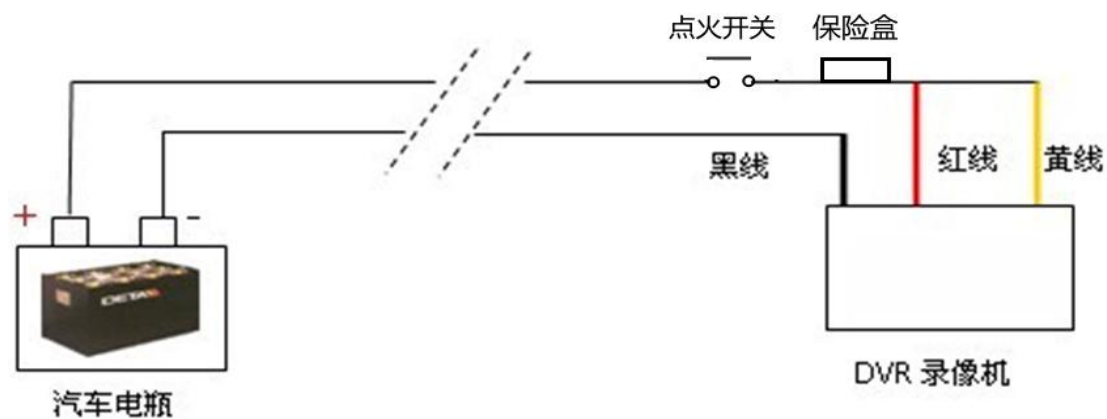
- a) Ensure that passengers or drivers cannot interfere with and damage equipment components, monitoring mainframes, cables, and other accessories, and do not install the equipment in close proximity to other restricted vehicle components.
- b) When installing the equipment components, monitoring terminal, accessories and cables, starting the vehicle may cause damage to the equipment, make sure the vehicle is stationary during installation to prevent the equipment from falling.
- c) Before removing and installing the SIM card, please make sure that the monitoring terminal is in the power-off state, otherwise it is easy to burn the card.
- d) Before unplugging the hard disc of the monitoring terminal, please make sure that the monitoring host is in a power-off state, otherwise it is easy to cause the hard disc to be damaged.

2.3、Power cord connection

The monitoring host uses DC power input, and the normal operating range of input voltage is +9V ~ +36V. The power connection diagram is as follows: Wiring Mode 1:



Wiring Mode 2:



2.4、Hard drive loading and precautions

2.4.1 caveat

- Please use the recommended model of 2.5-inch hard drive and buy it from regular channels to ensure the quality of the drive;
- Disconnect the power supply of the main unit before installing or extracting the hard discs;
- Please make sure that the HDD is inserted in the right direction, with the front of the HDD facing upwards and the SATA connector facing inwards.
- Should be gently pressed into the hard drive cover, if it is stuck and can not be pressed in, you should promptly check the size of the hard drive or insertion direction is correct;
- Hard drive installation should avoid bumping, brute force installation, please

pick up and put down gently;

- After loading, please close the hard drive cover and lock it in time to prevent the hard disc from falling off during the vehicle driving process, and the device will not record when it is not locked;

2.4.2 Installation

- The installation method is as follows:

1) Lift the hard drive cover by gently wiggling it outwards with your finger.



2) Install the 2.5 inches hard drive into the HDD cage in the direction shown in the picture.



3) Snap back the hard drive cover.



4) Cover back the main lid and lock it with the key.



2.5、SD card and SIM card loading and cautions

2.5.1 caveat

- Disconnect the power supply of the device before installing and extracting the SD card and SIM card.;
- Please confirm the direction of SD card and SIM card insertion (as shown in the picture);
- SD card and SIM card should be gently pressed in, if there is stuck and can not be pressed in, you should promptly check whether the SD card and SIM card insertion direction is correct.
- After loading, please close the front cover of the device in time and lock it, the unit will not record if it is not locked.;

2.5.2 Installation method (as shown)



2.6、Device Configuration Enabling Functions

The contents of the host configuration are shown below and can be updated via USB drive. Please contact our technical staff for detailed configuration contents.



3、Fault resolution

1、Device does not start

- Check the input power supply of the equipment, check whether the power cord is wired correctly, whether the earth wire is connected back to the battery, and whether the insurance on the power cord is intact;
- Check that the input voltage of the power supply of the input device is higher than 12 V, that the supplied device current is satisfied, and that there is voltage on the ACC signal line (the same as the power supply voltage);

2、The device is always rebooting

- Check if the equipment supply voltage is insufficient and does not reach the start-up voltage of the equipment, causing the equipment to restart all the time;
- Incomplete KEY file used when calibrating binocular passenger flow, causing the device to keep rebooting;

3、No video recording of equipment

- Check whether the storage device is installed, whether the contact of the storage media is intact, and whether the storage media can be recognized normally by the computer;
- Check whether the camera has a video signal input to the device, and see whether there is a video screen in the operation screen;

4、 Video files without sound

- Check if the camera has audio capture;
- Go to the video channel settings and check that the audio is turned on;
- The channel where the recording function is realized must have a video input and record normally;

5、 Device dialing unsuccessful

- Check that the installed SIM card is recognized, that the SIM card is installed in the SIM1 position, and that the communication module is recognized;
- If the SIM card is recognized and the signal value is normal, if the signal value is too low, check whether the communication antenna is connected properly or the antenna position is changed;
- Check whether the SIM card is in normal use, not in arrears, with the data service function enabled, to ensure that the SIM card can be used normally to access the Internet;

6、 Abnormal equipment positioning

- Check that the positioning antenna is correctly installed with the receiving side facing up, and that there is a silk-screened "Positioning" logo on the positioning antenna mount at the back of the main unit;
- Installation position inside windscreen, check window film for special materials, Interference or not;
- Compare satellite signals in the same location using the mobile phone app "AndroiTS GPS Test";
- Check whether the antenna receiver head is blocked, the antenna receiver head can not be covered by debris, covered may not receive the signal;
- Check antenna for broken shorts;